
From RANS Prediction to Hydrofoil Optimization: A Benchmark of Neural Surrogates

Anonymous Author(s)
Affiliation
Address
email

Abstract

1 Neural flow surrogates are usually compared by field error, although their
2 practical value often lies in the designs they select. We compare CNN-
3 U-Net, a physics-regularized point network, FNO, and DeepONet on 381
4 OpenFOAM simulations of hydrofoils in water, using 64 cases from two
5 unseen NACA families for validation. The models predict RANS fields, lift,
6 drag, and pressure-based cavitation-inception risk, and then drive the same
7 four-parameter hydrofoil shape-optimization loop. The physics-regularized
8 network gives the best pressure accuracy ($R^2 = 0.966$), cavitation F1 score
9 (0.921), and inference speed; DeepONet gives the most accurate force coef-
10 ficients. Inference is 917–1801 times faster than OpenFOAM. Fresh Open-
11 FOAM simulations confirm L/D gains of 2.3–79.7% for the selected designs,
12 while a matched direct-CFD search reaches $L/D = 29.89$. Three surrogates
13 select CFD-verified designs above that local-search reference, with the best
14 reaching 45.96. Field accuracy, force accuracy, and optimization perfor-
15 mance are related but not interchangeable.

16 1 Introduction

17 For a hydrofoil operating in water, shape design is an optimization problem wrapped around
18 a flow solver. Each change in camber, thickness, or incidence requires another RANS solu-
19 tion, so even a modest search can consume hundreds of CFD evaluations; neural surrogates
20 move most of that search off the solver by learning the map from geometry and operating
21 conditions to flow and forces [Brunton et al., 2020]. The difficult question is not simply
22 which network minimizes field error, but which one preserves the quantities that determine
23 a useful design.

24 CNN-U-Nets emphasize local multiscale structure [Ronneberger et al., 2015]; FNO mixes
25 information globally in Fourier space [Li et al., 2021, Kovachki et al., 2023]; DeepONet
26 separates parameter and coordinate representations [Lu et al., 2021]; and physics-informed
27 models constrain learning with differential equations [Raissi et al., 2019]. Their relative
28 merits may therefore change as the target moves from pressure fields to forces and finally
29 to shape optimization.

30 Airfoil-surrogate research has progressed from field prediction [Guo et al., 2016, Bhatnagar
31 et al., 2019, Sekar et al., 2019, Thuerey et al., 2020] to benchmarks that evaluate surface
32 forces, out-of-distribution accuracy, and computational cost [Bonnet et al., 2022, Li et al.,
33 2023, Yagoubi et al., 2025]. Those metrics establish whether a model approximates CFD,
34 but they do not by themselves establish whether it will make the same design decision.
35 A small, structured force error can be amplified when an optimizer repeatedly queries a
36 narrow region of design space. The missing experiment is therefore a controlled comparison

that follows several surrogate families through the same optimizer and then checks the selected geometries with new CFD solutions. High-fidelity endpoint verification is standard in surrogate shape optimization, while a matched direct-CFD search provides a stronger reference when its cost is manageable [Li et al., 2022, Shukla et al., 2023].

That experiment is the contribution of this paper; we do not propose a new architecture. We train four representative model families on the same 381 RANS simulations and reserve two complete NACA families for validation. We compare fields, total forces, pressure-threshold cavitation screening, and runtime, then give every model the same three-start hydrodynamic shape optimization (HSO) problem. Fresh OpenFOAM solutions check the four surrogate winners, while a direct OpenFOAM search uses the same starts and budget as a reference. Cavitation risk here means inception inferred from C_p , not a multiphase prediction of cavity evolution.

2 Data and CFD Reference

2.1 Design space

The dataset contains NACA 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization Abbott and von Doenhoff [1959]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. All cases use liquid water with density 997 kg/m^3 and kinematic viscosity $10^{-6} \text{ m}^2/\text{s}$. With a 1 m chord, the freestream speed is set by $U_\infty = Re\nu/c$; the far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM `simpleFoam` [Weller et al., 1998]. For steady incompressible flow, the mean velocity $\bar{u}_i$ and pressure $\bar{p}$ satisfy

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \rho \frac{\partial \overline{u'_i u'_j}}{\partial x_j}. \quad (2)$$

The Reynolds stress is closed with the k - ω SST model [Menter, 1994]. Angled freestream velocity is imposed at the inlet and far boundary, and OpenFOAM’s `forceCoeffs` supplies total lift and drag after 300 iterations. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA descriptors, a fluid mask, and signed distance to the profile.

Three simulations with $|C_L|$ or $|C_D| > 5$ are discarded before splitting, leaving 381 cases. All 64 cases from NACA 0015 and NACA 2412 form the validation set; the remaining 317 cases train the models. Because complete families are reserved, neither validation geometry appears in training at a different operating condition. All retained solves reach iteration 300. The 95th-percentile maximum initial residual at that iteration is 1.27×10^{-4} ; the corresponding late-window spans are 0.0059 in C_L and 2.65×10^{-4} in C_D .

2.2 Targets and cavitation-risk definition

The common output vector contains U_x , U_y , absolute pressure p , eddy viscosity, k , ω , C_p , cavitation margin, C_L , and C_D . Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM’s kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \quad C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}. \quad (3)$$

For an ambient-pressure value p_a , the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y) \frac{1}{2} \rho U_\infty^2 - p_v. \quad (4)$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

3 Surrogates and Optimization

3.1 Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, GELU activations, and base width 16. The FNO has width 32, four operator blocks, and 12 retained Fourier modes per direction Ronneberger et al. [2015], Li et al. [2021]:

$$h_{l+1} = \sigma(W_l h_l + \mathcal{F}^{-1}(R_l \mathcal{F}(h_l))) . \quad (5)$$

The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. Its output has the form

$$\hat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x}) / \sqrt{64} + c_k. \quad (6)$$

following Lu et al. [2021]. The physics-regularized point network has three tanh layers of width 64 and uses 1,024 sampled points per case. Its supervised loss is augmented by 10^{-4} times the continuity and first-order momentum residual loss Raissi et al. [2019]. This is a lightweight physics regularizer, not a data-free PINN or complete RANS residual.

All models use AdamW with learning rate 2×10^{-3} , weight decay 10^{-4} , at most 120 epochs, patience 20, gradient clipping, and seed 7. Normalization uses training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table 1).

Optimization uses dedicated C_L and C_D output heads supervised by OpenFOAM’s total force coefficients. Pressure integration on the coarse common grid is not used for the objective because it omits viscous traction and under-resolves the wall.

3.2 Common optimization protocol

At $Re = 5 \times 10^5$, each surrogate drives the same HSO loop. The design vector $\theta = (m, p, t, \alpha)$ contains maximum camber, camber location, thickness, and incidence, with bounds $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. We launch Nelder–Mead [Nelder and Mead, 1965] from NACA 0012, 2412, and 4415 and give every run the same 40-iteration budget. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}} - 10P_M - 5f_M, \quad (7)$$

where P_M penalizes a predicted minimum cavitation margin below 5000 Pa and f_M is the fraction of cells below that margin. The search calls only the surrogate; OpenFOAM is not used inside the loop. Since the nominal-pressure margin stays above the threshold for every selected design, the penalty is inactive in this experiment. We return the best design observed within the fixed budget, select the strongest of the three starts for each architecture, and then generate a new mesh and independent OpenFOAM simulation for that winner.

The direct-CFD baseline repeats this protocol with the same starts, bounds, objective, tolerances, and 40-iteration budget, but OpenFOAM supplies every objective value. It is distinct from the one-solve endpoint checks of the surrogate winners.

4 Results

4.1 Validation accuracy and computational cost

Table 1 reports results on the 64-case validation set. The physics-regularized network is best for pressure and C_p , with pressure RMSE 6.78 Pa and $R^2 = 0.966$. DeepONet gives the best U_x , U_y , C_L , and C_D coefficients. The grid operators are weaker in this low-data

Table 1: Validation performance, model size, and cost. Force RMSE is computed case-wise against OpenFOAM total coefficients.

Model	p RMSE	p R^2	C_L RMSE	C_D RMSE	Params.	Speedup
CNN-U-Net	28.52	0.392	0.0991	0.00453	485,178	1,148×
Physics-reg. MLP	6.78	0.966	0.0245	0.00204	9,930	1,801×
FNO	22.73	0.614	0.1383	0.00957	595,914	1,028×
DeepONet	8.55	0.945	0.0268	0.00184	100,746	917×

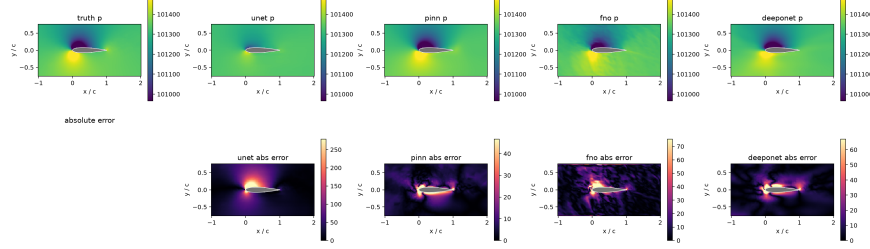


Figure 1: Absolute-pressure prediction for a representative validation case from the NACA 0015 case (case 160), followed by per-model absolute error. All prediction panels share the truth color scale; error panels use individually labeled scales.

setting. FNO retains useful force trends, while CNN-U-Net predicts C_p and forces better than absolute pressure or streamwise velocity.

Measured training times are 868, 104, 547, and 54 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is 12.4 s per case, while pressure-field inference requires 0.0069–0.0135 s, yielding 917–1801× speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on CPU and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

Figure 1 shows a representative validation case from the unseen NACA 0015 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. The distinct error color scales in the lower row are retained to expose each model’s error structure rather than visually equalizing models with very different ranges.

4.2 Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so accuracy is not useful here. The physics-regularized network achieves precision 0.927, recall 0.916, and F1 0.921. DeepONet follows at F1 0.913. CNN-U-Net has the highest recall (0.958) but lower precision (0.796), while FNO obtains F1 0.771. Twenty-four validation case/ambient-pressure combinations contain at-risk cells. This measures pressure-based inception screening, not cavity dynamics. Multiphase labels are required to predict vapor fraction, cavity length, or shedding Brennen [1995].

4.3 Shape optimization against direct CFD

Figure 2 reports the complete L/D comparison: the OpenFOAM value of the starting foil, the surrogate estimate at its selected design, and the result of remeshing that design and solving it once in OpenFOAM. This one-solve endpoint check is distinct from the direct baseline, which places OpenFOAM inside the matched optimizer.

All four surrogate winners improve on their starts. The physics-regularized network moves from $L/D = 25.57$ to 45.96, an absolute gain of 20.39 (79.7%); DeepONet gains 14.97, CNN-U-Net gains 12.55, and FNO gains 0.58. Direct CFD reaches 17.80, 27.33, and

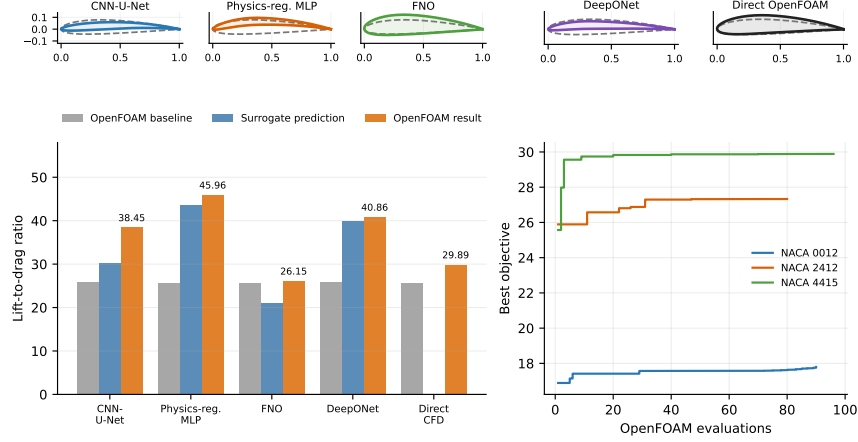


Figure 2: Top: selected shapes against the same dashed NACA 2412 geometric reference. The reference is for visual comparison, not necessarily the optimizer initializer. Bottom left: surrogate predictions, CFD endpoint checks, and direct CFD. Bottom right: direct-CFD convergence from all three initializers.

29.89 from NACA 0012, 2412, and 4415, respectively, using 266 evaluations in total. Its best gain is 4.32 (16.9%). The selected (m, p, t, α) values are $(0.0325, 0.550, 0.060, 3.06^\circ)$, $(0.0667, 0.388, 0.060, 1.24^\circ)$, $(0.0391, 0.372, 0.168, 4.23^\circ)$, and $(0.0335, 0.328, 0.060, 1.60^\circ)$ for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively; direct CFD selects $(0.0407, 0.417, 0.144, 3.98^\circ)$.

5 Discussion

Accuracy and decision quality. The results expose two distinct error modes. DeepONet has the lowest endpoint L/D error (2.1%), matching its strong force metrics, yet the physics-regularized network selects the better design: $L/D = 45.96$ rather than 40.86. CNN-U-Net and FNO underpredict by 21.5% and 19.3%, but only CNN-U-Net locates a large gain. Value calibration and candidate ranking are therefore separate capabilities; endpoint CFD changes the model comparison rather than merely confirming the force-error table.

The matched direct search reaches only $L/D = 29.89$. CNN-U-Net, DeepONet, and the physics-regularized model exceed it by 8.56, 10.97, and 16.07 after CFD validation, while FNO remains below it. The direct run is not a global optimum; it shows that learned objectives can lead the same budgeted local optimizer to different basins.

Search behavior. Initialization matters. The physics-regularized network reaches predicted $L/D = 43.55$ and 43.62 from the two cambered starts but only 22.35 from NACA 0012; FNO stays between 16.18 and 21.11. The three stronger designs approach 6% thickness, whereas FNO selects 16.8%. The pointwise models thus learn a sharper preference for thin, cambered profiles in this region.

Cost and cavitation. Across the three starts, the models require 199–249 evaluations, approximately 41–52 minutes if replaced one-for-one by the measured CFD solve versus 1.4–3.0 s of surrogate inference. The measured direct baseline uses 266 CFD evaluations and 38.5 minutes of summed wall time; each surrogate additionally requires one endpoint solve.

All selected designs have a nominal predicted cavitation margin near 99 kPa, so the penalty in Eq. (7) is inactive. The cavitation results establish screening accuracy, not a cavitation-shaped optimum; demonstrating that tradeoff requires reduced ambient pressure or multi-phase labels. The remaining scope is one seed, steady 2D RANS, and the NACA four-digit family; grid convergence and external force validation remain future checks.

6 Conclusion

We followed four neural surrogates through the complete path from RANS-field prediction to a design decision. On unseen NACA families, the physics-regularized point network gives the best pressure result ($R^2 = 0.966$) and cavitation-inception F1 score (0.921), while DeepONet gives the most accurate lift and drag. Every model evaluates a pressure field in 6.9–13.5 ms, corresponding to 917–1801 $\times$ speedup over the measured OpenFOAM runs.

The HSO experiment changes the ranking from a simple accuracy table. DeepONet predicts its winner most accurately, but the physics-regularized network finds the best design, increasing OpenFOAM L/D from 25.57 to 45.96. CNN-U-Net and DeepONet also find substantial gains, from 25.89 to 38.45 and 40.86, respectively, whereas FNO reaches only 26.15 from 25.57. The direct-CFD search reaches 29.89, confirming that evaluator choice changes which basin a limited local search reaches. A useful scientific surrogate should therefore be judged on both its validation predictions and the consequence of acting on them.

References

- Ira H. Abbott and Albert E. von Doenhoff. Theory of Wing Sections. Dover Publications, 1959.
- Saakaar Bhatnagar, Yaser Afshar, Shaowu Pan, Karthik Duraisamy, and Shailendra Kaushik. Prediction of aerodynamic flow fields using convolutional neural networks. *Computational Mechanics*, 64(2):525–545, 2019. doi: 10.1007/s00466-019-01740-0.
- Florent Bonnet, Jocelyn Ahmed Mazari, Paola Cinnella, and Patrick Gallinari. AIRFRANS: High fidelity computational fluid dynamics dataset for approximating reynolds-averaged navier–stokes solutions. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Christopher E. Brennen. *Cavitation and Bubble Dynamics*. Oxford University Press, 1995.
- Steven L. Brunton, Bernd R. Noack, and Petros Koumoutsakos. Machine learning for fluid mechanics. *Annual Review of Fluid Mechanics*, 52:477–508, 2020. doi: 10.1146/annurev-fluid-010719-060214.
- Xiaoxiao Guo, Wei Li, and Francesco Iorio. Convolutional neural networks for steady flow approximation. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 481–490, 2016. doi: 10.1145/2939672.2939738.
- Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- Jichao Li, Xiaosong Du, and Joaquim R. R. A. Martins. Machine learning in aerodynamic shape optimization. *arXiv preprint arXiv:2202.07141*, 2022.
- Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- Zongyi Li, Nikola Kovachki, Chris Choy, Boyi Li, Jean Kossaifi, Shourya Otta, Mohammad Amin Nabian, Maximilian Stadler, Christian Hundt, Kamyar Azizzadenesheli, and Animashree Anandkumar. Geometry-informed neural operator for large-scale 3d PDEs. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021. doi: 10.1038/s42256-021-00302-5.

225 Florian R. Menter. Two-equation eddy-viscosity turbulence models for engineering applica-
 226 tions. *AIAA Journal*, 32(8):1598–1605, 1994. doi: 10.2514/3.12149.

227 John A. Nelder and Roger Mead. A simplex method for function minimization. *The Com-*
 228 *puter Journal*, 7(4):308–313, 1965. doi: 10.1093/comjnl/7.4.308.

229 Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural net-
 230 works: A deep learning framework for solving forward and inverse problems involving
 231 nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707,
 232 2019. doi: 10.1016/j.jcp.2018.10.045.

233 Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for
 234 biomedical image segmentation. In *Medical Image Computing and Computer-Assisted*
 235 *Intervention*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.

236 Vinothkumar Sekar, Qinghua Jiang, Chuan Shu, and Boo Cheong Khoo. Fast flow field
 237 prediction over airfoils using deep learning approach. *Physics of Fluids*, 31(5):057103,
 238 2019. doi: 10.1063/1.5094943.

239 Khemraj Shukla, Vivek Oommen, Ahmad Peyvan, Michael Penwarden, Luis Bravo, Anindya
 240 Ghoshal, Robert M. Kirby, and George Em Karniadakis. Deep neural operators can serve
 241 as accurate surrogates for shape optimization: A case study for airfoils. *arXiv preprint*
 242 *arXiv:2302.00807*, 2023.

243 Nils Thuerey, Konstantin Weissenow, Lukas Prantl, and Xiangyu Hu. Deep learning meth-
 244 ods for reynolds-averaged navier–stokes simulations of airfoil flows. *AIAA Journal*, 58(1):
 245 25–36, 2020. doi: 10.2514/1.J058291.

246 Henry G. Weller, G. Tabor, Hrvoje Jasak, and C. Fureby. A tensorial approach to computa-
 247 tional continuum mechanics using object-oriented techniques. *Computers in Physics*, 12
 248 (6):620–631, 1998. doi: 10.1063/1.168744.

249 Mouadh Yagoubi, David Danan, Milad Leyli-Abadi, Jocelyn Mazari, Jean-Patrick Brunet,
 250 Abbas Kabalan, Fabien Casenave, Yuxin Ma, Giovanni Catalani, Jean Fesquet, Jacob
 251 Helwig, Xuan Zhang, Haiyang Yu, Xavier Bertrand, Frederic Tost, Michael Bauerheim,
 252 Joseph Morlier, and Shuiwang Ji. ML4CFD competition: Results and retrospective anal-
 253 ysis. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

254 NeurIPS Paper Checklist

255 1. Claims

256 Question: Do the main claims made in the abstract and introduction accurately
 257 reflect the paper’s contributions and scope?

258 Answer: [\[Yes\]](#)

259 Justification: The abstract, introduction, and limitations distinguish CFD agree-
 260 ment, physical validation, and pressure-based inception screening.

261 Guidelines:

- 262 • The answer [\[N/A\]](#) means that the abstract and introduction do not include the
 263 claims made in the paper.
- 264 • The abstract and/or introduction should clearly state the claims made, in-
 265 cluding the contributions made in the paper and important assumptions and
 266 limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well
 267 by the reviewers.
- 268 • The claims made should match theoretical and experimental results, and reflect
 269 how much the results can be expected to generalize to other settings.
- 270 • It is fine to include aspirational goals as motivation as long as it is clear that
 271 these goals are not attained by the paper.

272 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 discusses simulator fidelity, the single training seed, geometry scope, direct force supervision, and limited CFD revalidation.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper presents an empirical benchmark and no theoretical results or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 2–3 specify the case matrix, split, CFD setup, model families, losses, hyperparameters, and optimization protocol.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The repository is organized for release, but an anonymized archival URL has not yet been inserted in this draft.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.

- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 2–3 give the deterministic split, seed, optimizer, learning rate, model sizes, stopping rule, and design bounds.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Models were trained with one fixed seed; the paper reports per-case held-out metrics and explicitly identifies repeated-seed analysis as future work.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 4 reports CPU hardware, memory, software versions, training time, solver time, inference time, and amortized break-even counts.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research uses generated CFD data, involves no people or sensitive data, and preserves limitations and exclusions.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 5 discusses computational benefits and the risk that surrogate speed can scale simulator bias or invalid cavitation claims.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The generated hydrofoil models and CFD fields do not present a high-risk dual-use release.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: OpenFOAM and prior architectures are cited, but explicit software-license text remains to be added to the artifact documentation.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/

538 **datasets** has curated licenses for some datasets. Their licensing guide can help
539 determine the license of a dataset.

- 540 • For existing datasets that are re-packaged, both the original license and the
541 license of the derived asset (if it has changed) should be provided.
- 542 • If this information is not available online, the authors are encouraged to reach
543 out to the asset’s creators.

544 13. New assets

545 Question: Are new assets introduced in the paper well documented and is the
546 documentation provided alongside the assets?

547 Answer: [No]

548 Justification: The generated dataset, checkpoints, and scripts are documented lo-
549 cally, but the anonymized release and final license are not yet attached to this
550 draft.

551 Guidelines:

- 552 • The answer [N/A] means that the paper does not release new assets.
- 553 • Researchers should communicate the details of the dataset/code/model as part
554 of their submissions via structured templates. This includes details about train-
555 ing, license, limitations, etc.
- 556 • The paper should discuss whether and how consent was obtained from people
557 whose asset is used.
- 558 • At submission time, remember to anonymize your assets (if applicable). You
559 can either create an anonymized URL or include an anonymized zip file.

560 14. Crowdsourcing and research with human subjects

561 Question: For crowdsourcing experiments and research with human subjects, does
562 the paper include the full text of instructions given to participants and screenshots,
563 if applicable, as well as details about compensation (if any)?

564 Answer: [N/A]

565 Justification: The work contains no crowdsourcing or human-subject research.

566 Guidelines:

- 567 • The answer [N/A] means that the paper does not involve crowdsourcing nor
568 research with human subjects.
- 569 • Including this information in the supplemental material is fine, but if the main
570 contribution of the paper involves human subjects, then as much detail as
571 possible should be included in the main paper.
- 572 • According to the NeurIPS Code of Ethics, workers involved in data collection,
573 curation, or other labor should be paid at least the minimum wage in the
574 country of the data collector.

575 15. Institutional review board (IRB) approvals or equivalent for research with human
576 subjects

577 Question: Does the paper describe potential risks incurred by study participants,
578 whether such risks were disclosed to the subjects, and whether Institutional Review
579 Board (IRB) approvals (or an equivalent approval/review based on the requirements
580 of your country or institution) were obtained?

581 Answer: [N/A]

582 Justification: The work contains no human-subject research requiring IRB review.

583 Guidelines:

- 584 • The answer [N/A] means that the paper does not involve crowdsourcing nor
585 research with human subjects.
- 586 • Depending on the country in which research is conducted, IRB approval (or
587 equivalent) may be required for any human subjects research. If you obtained
588 IRB approval, you should clearly state this in the paper.

589 • We recognize that the procedures for this may vary significantly between insti-
590 tutions and locations, and we expect authors to adhere to the NeurIPS Code
591 of Ethics and the guidelines for their institution.

592 • For initial submissions, do not include any information that would break
593 anonymity (if applicable), such as the institution conducting the review.

594 16. Declaration of LLM usage

595 Question: Does the paper describe the usage of LLMs if it is an important, original,
596 or non-standard component of the core methods in this research? Note that if the
597 LLM is used only for writing, editing, or formatting purposes and does not impact
598 the core methodology, scientific rigor, or originality of the research, declaration is
599 not required.

600 Answer: [N/A]

601 Justification: LLMs are not part of the scientific method, model architectures, data
602 generation, or evaluation reported in the paper.

603 Guidelines:

604 • The answer [N/A] means that the core method development in this research
605 does not involve LLMs as any important, original, or non-standard components.

606 • Please refer to our LLM policy in the NeurIPS handbook for what should or
607 should not be described.